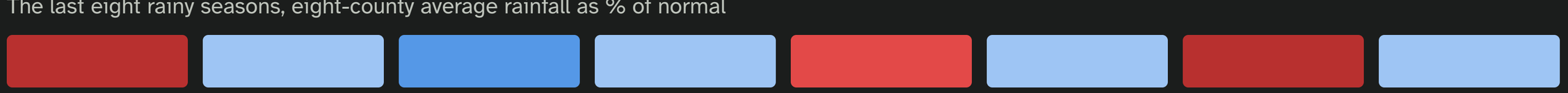
DROUGHT · HEAVY RAIN · HEAT · NORTHERN KENYA

Ukame Watch

Early warning for eight arid counties, built so a county officer can act before a season fails

The last eight rainy seasons, eight-county average rainfall as % of normal



43% Oct–Dec 22 | **133%** Mar–May 23 | **174%** Oct–Dec 23 | **136%** Mar–May 24 | **70%** Oct–Dec 24 | **111%** Mar–May 25 | **42%** Oct–Dec 25 | **129%** Mar–May 26

Zetech Alumni × Student Hackathon 2026 · ML & Data Science · Climate Sentinel

Patrick Wangai · Faith Githaiga · Monchari Leah · Dedan Mugambi

SPEAKER NOTES · SLIDE 1 OF 10

Open with the strip, not the title. These are the last eight rainy seasons across our eight counties: 43 percent of normal, then 174, then 70, then 42. Failure, flood, failure. That whiplash is the problem. Ukame is Swahili for drought. Ukame Watch tells a county disaster officer, in plain language, whether this season is unusual, what is coming, who is exposed and what to prepare, early enough to act. About 20 seconds on this slide.

REAL-WORLD PROBLEM AND SECTOR RELEVANCE

Drought, then flood, in the same counties

5.6 million

people in Turkana, Marsabit, Wajir, Mandera, Garissa, Isiolo, Samburu and Kitui

KNBS 2019 census

5 droughts

documented since 2000; three were declared national disasters

FAO, World Weather Attribution, ACAPS

135,125

people displaced by the late-2023 floods; Garissa, Mandera and Wajir worst hit

UN OCHA, 20 November 2023

Before a season fails, a county disaster officer needs four answers: **Is this season unusual? What comes next? Who is exposed? What do we prepare?**

SPEAKER NOTES · SLIDE 2 OF 10

The users are county disaster managers and the NDMA officers they work with. Five major droughts since 2000, three declared national disasters, in January 2006, February 2017 and September 2021. Then the opposite: the late-2023 floods displaced over 135,000 people, with Garissa, Mandera and Wajir worst hit. Officers have NDMA bulletins and KMD forecasts, but no single place that answers their four questions per county. That is the gap. About 40 seconds.

SOLUTION RELEVANCE, INNOVATION AND SUITABILITY

One board answers the four questions

Is this season unusual?

Rainfall as % of the 1991–2020 normal, and an NDMA-style drought level for each county

What comes next?

A 3-month outlook tested against two baselines, and a live 7-day heavy-rain and heat watch

Who is exposed?

Census population per county, and the trend in hot days since 1991

What do we prepare?

A cited action playbook, and an offline AI that drafts an advisory a named officer must approve

WHAT IS NEW

A forecast that reports where it fails, and an AI that cannot invent a number, skip the playbook or bypass a person

SPEAKER NOTES · SLIDE 3 OF 10

Each of the four questions maps to one part of the product. The innovation is not another dashboard: it is honesty built into the system. The forecast tells you where it has no skill. The AI runs offline on a laptop, uses six tools, and code checks every number against those tools, blocks claims like no risk, adds the cited playbook actions itself, and holds every advisory as a draft until a named person approves. It suits counties with weak connectivity and no budget. About 45 seconds.

LIVE TODAY · 13 NOVEMBER 2026

Samburu is at drought Alarm; two counties face heavy rain

26% of normal rain

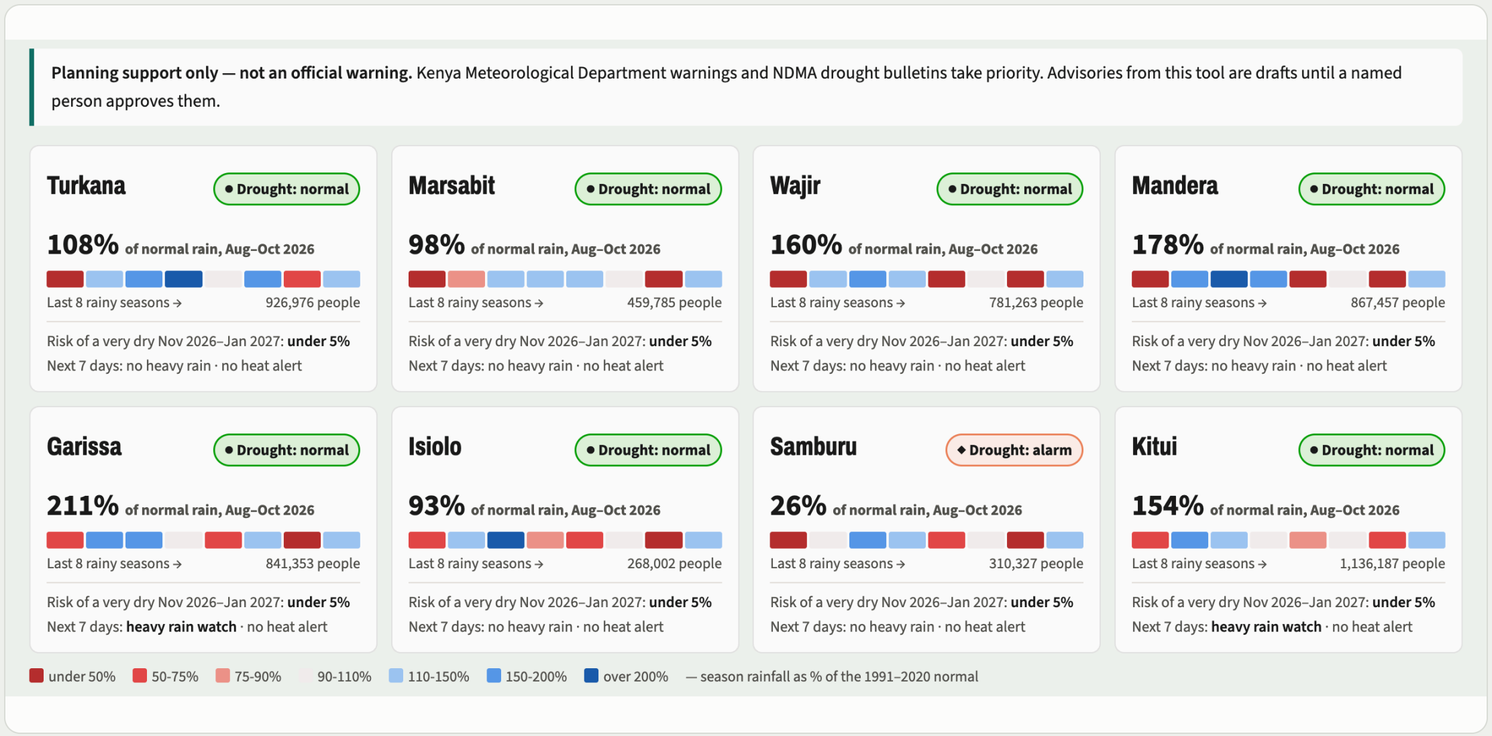
Samburu, Aug–Oct 2026: drought level Alarm

Heavy-rain watch

Next 7 days: Garissa and Kitui

Under 5%

risk of a very dry Nov–Jan, with a strong El Niño (+2.46) and positive Indian Ocean Dipole (+0.95)



SPEAKER NOTES · SLIDE 4 OF 10

Switch to the live site here if time allows, otherwise talk over the screenshot. Samburu has had 26 percent of normal rain over August to October, two windows in a row below 75 percent, so the board shows Alarm. At the same time the live forecast puts two counties, Garissa and Kitui, on a heavy-rain watch. Both are true at once: that is the whiplash from the cover. BEFORE PRESENTING: open the live site the morning of the finale and update the county list on this slide, because the 7-day watch changes daily. About 45 seconds.

MEASURABLE OUTCOMES AND EVALUATION

It catches the disasters Kenya already lived through

10 of 10 documented droughts and floods, 1997–2024, flagged by our indicators

6 of 6 worst droughts and heavy-rain periods in 35 years of data match a documented disaster

+0.12 forecast skill for Oct–Dec rains on unseen years 2017–2026 (Brier skill score vs climatology); persistence scores −0.01

+5 to +10 extra hot days per decade, statistically significant in 7 of 8 counties

Where it fails: no skill for the Mar–May long rains (−0.05). The app says so on screen.

3 of the 10 events were added after the indicator flagged them, then verified independently; they are labelled.

SPEAKER NOTES · SLIDE 5 OF 10

This is the slide that earns trust. We checked our indicators against ten documented disasters from 1997 to 2024 and caught all ten. The six worst periods in 35 years of data all match real events. The forecast was trained on 1991 to 2016 and tested on years it never saw: for the October to December rains it beats climatology and persistence. For March to May it does not, and we tell users that. Honest: three events were added after our indicator flagged them, then checked against FAO and Red Cross sources, and we label them. About 45 seconds.

TECHNICAL ARCHITECTURE AND TECHNOLOGY STACK

From free public data to an approved advisory

Free data
NASA POWER daily since 1991, ERA5, NOAA El Niño and Indian Ocean indices, census

→

Quality control
11 impossible rainfall days caught by cross-checking a second dataset

→

Indicators and forecast
Drought, heavy rain and heat; one logistic model per season

→

Local AI with guardrails
Qwen 2.5 3B, six tools, every number checked in code

→

Person approves
Drafts stay drafts until a named officer approves; full audit log

93 automated tests · **One command rebuilds every number in about a minute** · **Runs offline on an 8 GB laptop** · **No GPU, no paid services**

Python · pandas · scikit-learn · pymannkendall · GeoPandas · Streamlit · Plotly · Ollama · stlite on GitHub Pages

SPEAKER NOTES · SLIDE 6 OF 10

Walk left to right. Free public data from NASA, ERA5 and NOAA. Quality control: our main source had a day showing 608 millimetres of rain in Turkana, when ERA5 showed one millimetre, so we cross-check and fixed eleven such days. Then the indicators and a simple, explainable forecast model. Then a small AI that runs locally and can only use six tools, with every number it writes checked in code. Finally a person approves. All of it is tested and rebuilds from scratch with one command. About 40 seconds.

RISK IDENTIFICATION, SAFETY AND SUSTAINABILITY

We designed for what can go wrong

Risk	What we built
Rainfall data is model-based	Cross-checked with ERA5; 11 impossible days found and fixed
The AI invents or overstates	Every number and level word checked; “no risk” blocked; a person approves
Forecasts look too certain	Below 5% shown as “under 5%”; Mar–May labelled no skill
Mistaken for official warnings	On every screen: KMD and NDMA warnings take priority
Wrong population figures	Census used; WorldPop is 3 times too high in Mandera
Running costs or servers fail	Free public data, free hosting, works offline on a laptop

Full list: docs/limitations.md in the repository

SPEAKER NOTES · SLIDE 7 OF 10

Six risks and what we did about each. The two that matter most: wrong data and an AI that makes things up. For data, we cross-check sources and log every correction; we also found that a popular population dataset overstates Mandera three times, so we use the census. For the AI, code checks every number and every drought level word, refuses cyclone-path questions and counties we do not cover, blocks claims like no risk, and nothing goes out without a named person approving it. In our demo video the code caught the AI writing Alert when the data said Alarm. We never present this as an official warning. About 40 seconds.

REAL-WORLD IMPLEMENTATION, FEASIBILITY AND SCALABILITY

Free to run today, ready to scale

Works today

- Zero running cost: free public data, free hosting
- Runs offline on an 8 GB laptop; nothing leaves the device
- One command rebuilds every number from scratch
- Open source, with 93 automated tests

Next 12 months

1. Pilot with one County Steering Group in the 2027 long rains
2. Add Kenya Meteorological Department station rainfall
3. Extend to every county NDMA monitors: settings plus validation
4. Swahili summaries by SMS and radio for pastoralists

SPEAKER NOTES · SLIDE 8 OF 10

Feasibility is where most prototypes fail, so we designed for a county office with no budget and weak internet. It costs nothing to run, works offline on a basic laptop, and anyone can rebuild every number with one command. Scaling is mostly configuration: the eight counties are a setting, so extending to every county NDMA monitors means adding names and re-validating. The first real step is a pilot with one County Steering Group, then adding KMD station data to correct the reanalysis, then Swahili summaries for pastoralists. About 40 seconds.

SDG ALIGNMENT AND SOCIETAL IMPACT

Earlier action against Kenya’s costliest climate hazards

SDG 13.1

Climate resilience: county signals before a season fails

SDG 2.4

Food security: time for water repairs and timely livestock offtake

SDG 1.5

Poverty: protect household assets in some of Kenya’s poorest counties

SDG 11.5

Disasters: heavy-rain watches prompt early warning and evacuation plans

HOW A PILOT WILL MEASURE IMPACT

Lead time: days from signal to county action

Trust: share of drafts approved without edits

Use: officer-rated usefulness each month

Also serves livestock and agriculture officers, and humanitarian partners such as the Kenya Red Cross

SPEAKER NOTES · SLIDE 9 OF 10

Primary goal is SDG 13, climate resilience, with direct links to zero hunger, poverty and disaster risk. Be honest about the claim: Ukame Watch shortens the time between a rainfall signal and a county conversation about preparedness. It does not measure hunger or losses, and we have not tested it with officers yet. So here is how a pilot would prove impact: lead time from signal to action, how often officers approve drafts unchanged, and how useful they rate it. About 35 seconds.

UKAME WATCH

Earlier warnings for 5.6 million people, and honest about their limits

Try it now: patrickwangai.github.io/ukame-watch

Code, data sources and limitations: github.com/PatrickWangai/ukame-watch

Patrick Wangai · Faith Githaiga · Monchari Leah · Dedan Mugambi · Thank you

SPEAKER NOTES · SLIDE 10 OF 10

Close on the one sentence: earlier warnings for 5.6 million people, honest about their limits. Invite the judges to open the live link on their phones; it runs in the browser. Then stop and take questions. Likely questions: why logistic regression (explainable, small data, it beats the baselines), why a 3B model (runs offline on county laptops, and code guards it), and what happens when it is wrong (a person approves every draft, and limitations are on every screen). About 20 seconds.